

Section 1: Principles of chemistry

- a) States of matter
- b) Atoms
- c) Atomic structure
- d) Relative formula masses and molar volumes of gases
- e) Chemical formulae and chemical equations
- f) Ionic compounds
- g) Covalent substances
- h) Metallic crystals
- i) Electrolysis

a) States of matter

Students will be assessed on their ability to:

- 1.1 understand the arrangement, movement and energy of the particles in each of the three states of matter: solid, liquid and gas
- 1.2 describe how the interconversion of solids, liquids and gases are achieved and recall the names used for these interconversions
- 1.3 describe the changes in arrangement, movement and energy of particles during these interconversions.

b) Atoms

Students will be assessed on their ability to:

- 1.4 describe simple experiments leading to the idea of the small size of particles and their movement including:
 - i dilution of coloured solutions
 - ii diffusion experiments
- 1.5 understand the terms atom and molecule
- 1.6 understand the differences between elements, compounds and mixtures
- 1.7 describe techniques for the separation of mixtures, including simple distillation, fractional distillation, filtration, crystallisation and paper chromatography.

c) Atomic structure

Students will be assessed on their ability to:

- 1.8 recall that atoms consist of a central nucleus, composed of protons and neutrons, surrounded by electrons, orbiting in shells
- 1.9 recall the relative mass and relative charge of a proton, neutron and electron
- 1.10 understand the terms atomic number, mass number, isotopes and relative atomic mass (A_r)
- 1.11 calculate the relative atomic mass of an element from the relative abundances of its isotopes
- 1.12 understand that the Periodic Table is an arrangement of elements in order of atomic number
- 1.13 deduce the electronic configurations of the first twenty elements from their positions in the Periodic Table
- 1.14 deduce the number of outer electrons in a main group element from its position in the Periodic Table.

d) Relative formula masses and molar volumes of gases

Students will be assessed on their ability to:

- 1.15 calculate relative formula masses (M_r) from relative atomic masses (A_r)
- 1.16 understand the use of the term mole to represent the amount of substance
- 1.17 understand the term mole as the Avogadro number of particles (atoms, molecules, formulae, ions or electrons) in a substance**
- 1.18 carry out mole calculations using relative atomic mass (A_r) and relative formula mass (M_r)